



```
1  module comparator (  
2  
3      input [3 : 0] A , [3 : 0] B  ,  
4      input EN ,  
5      output reg  equal , greater , less  
6  );  
7  
8      always @(*)  
9      begin  
10  
11          if(EN)  
12              begin  
13                  if(A>B) greater  =  1;  
14                  else if (A < B) less = 1;  
15                  else equal = 1;  
16              end  
17  
18          else  
19              begin  
20                  equal = 0 ;  
21                  greater = 0;  
22                  less =  0;  
23              end  
24      end  
25  endmodule
```

[illegible]



```
1  module alu (  
2  input [3 : 0] a , b ,  
3  input cin , pass_a , pass_b ,  
4  input [1 : 0] opcode ,  
5  
6  output reg [3 : 0] out ,  
7  output reg cout  
8  );  
9  
10 always @(*)  
11 begin  
12     if(pass_a) out  = a;  
13     else if(pass_b) out = b;  
14     else  
15         begin  
16             case(opcode)  
17  
18                 2'b00 :  out = a&b;  
19                 2'b01 :  {out , cout}=a+b+cin;  
20                 2'b10 :  out =  a - b;  
21                 2'b11 :  out  = {3'b000 , ^b}  ;  
22                 default : out  = 0 ;  
23             endcase  
24         end  
25     end  
26 endmodule
```

The screenshot displays the Logic Analyzer interface with the following components:

- Signal List (Left):** A list of signals including `a`, `b`, `cin`, `pass_a`, `pass_b`, `opcode`, `out`, and `cout`.
- Signal Details (Middle-Left):** A list of signals with their current values, all showing "-No Data-".
- Table (Middle-Right):** A table showing captured data for signals `a`, `b`, `cin`, `pass_a`, `pass_b`, `opcode`, `out`, and `cout`. The data is as follows:

Signal	Value
<code>a</code>	1001
<code>b</code>	0110
<code>cin</code>	
<code>pass_a</code>	01
<code>pass_b</code>	00
<code>opcode</code>	01
<code>out</code>	10
<code>cout</code>	
- Timing Diagram (Bottom):** A timing diagram showing the time progression of the signals. The time scale ranges from 0 ps to 600 ps. A red vertical line indicates the current time at 933 ps. A green horizontal line indicates the time interval from 0 ps to 933 ps.

Project Explorer

- + a
- + b
- cin
- pass_a
- pass_b
- + opcode
- + out
- cout

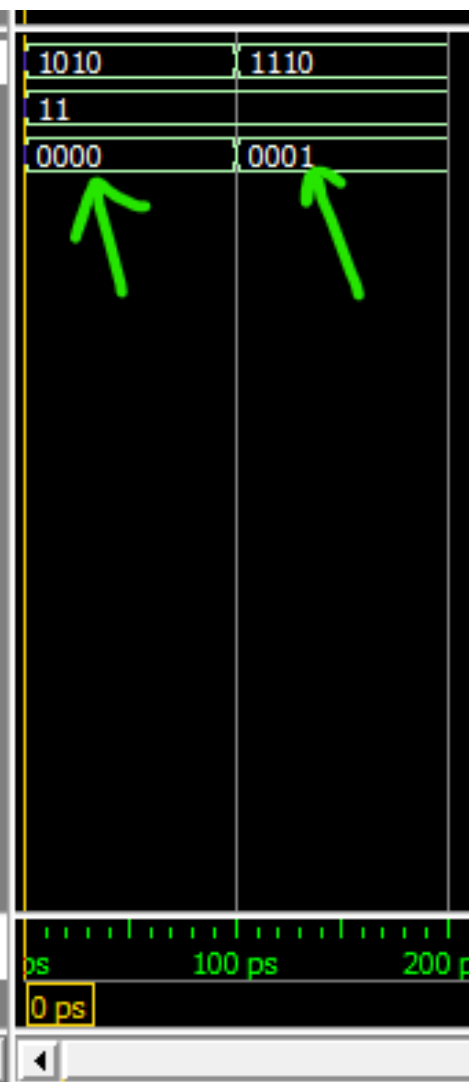
s (Active)

Name

Waveform Viewer

Signal	Value
/alu/b	1110
/alu/opcode	11
/alu/out	0001

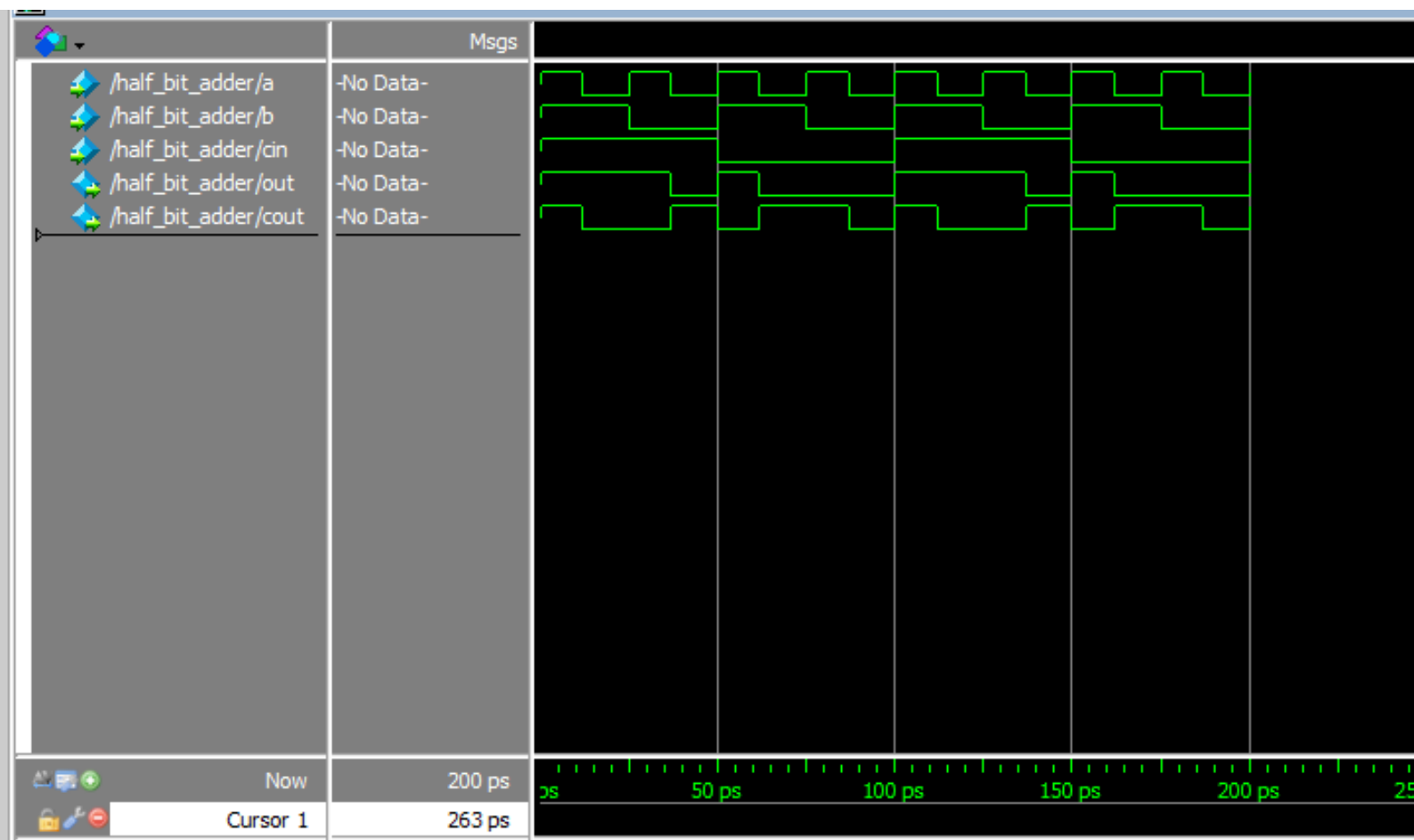
Now 200 ps
Cursor 1 0 ps



Find:



```
1  module half_bit_adder(  
2  
3  input a, b,cin,  
4  output out , cout  
5  
6  );  
7  assign {out,cout}=a+b+cin;  
8  endmodule
```





```
1  module full_2bit_adder (  
2  
3  input [1 : 0] a , b,  
4  input cin,  
5  output [1 : 0] sum,  
6  output cout  
7  );  
8  wire carry;  
9  full_1bit_adder f1(a[0],b[0],cin,sum[0],carry);  
10 full_1bit_adder f2(a[1],b[1],carry,sum[1],cout);  
11  
12 endmodule
```